

Exam Date & Time: 29-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

INTRODUCTION TO INTERNET OF THINGS [NPT-4016]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Build a disaster management IoT system that predicts and alerts authorities in real-time. Explain the roles of sensors and actuators in an IoT system. (5)
 - A)
 - B) If a chat application starts experiencing delays in message delivery when using REST APIs, Identify the limitations of this approach. (3)
 - C) List the IoT sensor devices to protect from cyberattacks in a smart home. (2)
- 2) Write an Arduino code for a home automation system where a button press controls a light and a fan. (5)
 - A)
 - B) In a smart home system, some sensors may use Zigbee to communicate, while others rely on Wi-Fi. If a Zigbee-enabled temperature sensor needs to trigger a Wi-Fi-enabled smart thermostat, a protocol translation gateway will convert the Zigbee message into a format compatible with the thermostat's Wi-Fi communication. Identify how protocol translation enhances interoperability in IoT systems and explain. (3)
 - C) You are building a weather monitoring system with Raspberry Pi and a DHT11 temperature and humidity sensor. Identify how the Raspberry Pi collects data from the DHT11 sensor, and what communication interface will be used. (2)
- 3) A smart greenhouse system uses Raspberry Pi to collect temperature, humidity, and light data from sensors. Analyze how data can be transmitted to a cloud platform for real-time monitoring. Include considerations such as communication protocols and data security. (5)
 - A)
 - B) Consider a smart building management system that utilizes IoT devices to monitor and analyze energy consumption. By analyzing historical IoT data, the building management team aims to identify trends and patterns that can inform operational decisions, optimize energy usage, and reduce costs. Apply the techniques to wring value from IoT data and explain. (3)
 - C) A small business is considering moving its operations to the cloud to reduce IT costs. Identify two advantages of using cloud computing that the business should consider in its decision-making process. (2)

- 4) Smart Water Management System (SWMS) implemented in a metropolitan area to monitor and manage water distribution and consumption. The system uses IoT devices to track water flow, detect and leaks. Analyze the impact of data latency on decision-making in IoT applications. (5)
- A)
- B) With the increasing urban population and vehicle density, predicting traffic congestion is crucial for improving traffic flow and reducing travel times. Identify a machine learning framework to predict traffic congestion based on real-time data collected from IoT devices. (3)
- C) Analyze a prototype for a mobile application that allows users to interact with and control multiple IoT devices remotely. What functionalities would you prioritize? (2)
- 5) Consider a patient with a chronic condition (e.g., diabetes) who is being monitored at home using various wearable devices. The goal is to collect real-time health data and transmit it to healthcare providers for analysis and decision-making. Analyze the data collection and data transmission for the given scenario. (5)
- A)
- B) A smart retail environment enhances the shopping experience through IoT technology by providing personalized services and efficient inventory management. Identify the IoT building blocks for the given scenario. (3)
- C) In a smart home, a fitness tracker uses a network to transmit data about the user's activity and heart rate to a smartphone app. Choose the suitable type of network used for this application and justify. (2)

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